

RITHIK KUMAR M B

Backend Developer | Java & Spring Boot

+91-9353543027 | ffrithik@gmail.com | github.com/Rithik333R | linkedin.com/in/rithik-kumar-m-b-117a571ba

SUMMARY

Backend developer focused on Java and Spring Boot, with experience building full-stack applications and integrating AI into real application workflows using Google Gemini. Built an AI-powered freelance management platform (Java, Spring Boot, React, PostgreSQL/pgvector, Docker) featuring semantic retrieval and a human-in-the-loop AI action system, and shipped a deployed AI-powered interview preparation platform. Computer Science graduate (2025) currently strengthening backend fundamentals and problem-solving skills, seeking a Java/Spring Boot backend developer role.

SKILLS

Languages: Java, Python, JavaScript, SQL

Backend: Spring Boot, Spring Security, REST APIs, JWT (with refresh tokens), Flyway

Frontend: React.js, JavaScript

Databases: PostgreSQL (pgvector), MySQL, MongoDB

AI & Integration: Google Gemini / Google GenAI, Text Embeddings, Semantic Retrieval, Apache PDFBox/POI, Cloudinary

Tools: Docker, Docker Compose, Git, GitHub, node-cron, Mongoose

EXPERIENCE

Software Engineer Intern (IoT) — STPI IoT Open Lab

Sep 2024 – Nov 2024

- Built the IoT integration and backend for a self-checkout smart cart system (small team project) using ESP32 to interface with RFID and load cell sensors — RFID identified the picked item, and load cell weight changes confirmed removal before the item was added to the cart.
- Owned the backend logic connecting sensor data to cart state, achieving 99% system uptime and reducing latency between hardware and backend systems.

Data Analyst Intern — Ants Skill Varsity

Feb 2025 – May 2025

- Cleaned and analyzed a COVID-19 dataset (10,000+ records) using Python (solo project).
- Built interactive Power BI dashboards to surface trends, reducing manual reporting time by 30%.

PROJECTS

AI Freelance Engineering OS

Feature-complete · Not deployed

Java, Spring Boot, React, PostgreSQL/pgvector, Docker, JWT, Google Gemini

- Full-stack freelance management platform with AI integrated into the workflow; Spring Boot + PostgreSQL/pgvector backend with JWT authentication and refresh tokens, role-based authorization, and Flyway migrations; React frontend with a persistent AI chat interface.
- AI layer uses Google Gemini for context-aware conversations, generating embeddings for clients/projects/tasks and retrieving relevant data by semantic similarity to ground responses.
- Designed a human-in-the-loop action system: AI proposes structured changes (create client/project/task) that require explicit user confirmation before execution; AI has no direct database access.
- Backend covered by 67 automated tests, including dedicated cross-user data isolation tests. Dockerized for local development.

AI Mock Interview Platform

Live — Vercel / Render / Aiven MySQL

React, Java/Spring Boot, MySQL, JWT, Google Gemini

- AI-powered interview preparation platform with resume upload (text extraction via Apache PDFBox/POI, file storage via Cloudinary) and difficulty-based mock interviews.
- Built a dedicated AI layer sending structured prompts to Google Gemini (Google AI Studio API) to generate interview questions, evaluate answers, produce an overall score, and calculate ATS compatibility and skill-gap analysis with a generated weekly learning roadmap.

Free Fire Arena

Live

Node.js, Express, MongoDB, JWT

- Tournament management platform for esports competitions across three modes — Battle Royale, Clash Squad, and Lone Wolf.
- JWT-based authentication with axios interceptors handling authorization; role-based access separating user and admin (Command Center) permissions.
- Automated tournament scheduling using node-cron; leaderboard rankings calculated via a MongoDB/Mongoose aggregation pipeline.

Employee Management System

Java, Spring Boot, JWT, MySQL

- Backend system for managing employee records with JWT authentication.

EDUCATION

B.E. Computer Science and Engineering — T John Institute of Technology, Bengaluru

Nov 2021 – Jul 2025

CGPA: 8.36 / 10

CERTIFICATIONS

IBM — Predictive Analytics Modeler (Oct 2024) • IBM Watson Studio (Feb 2024) • Cloud Application Developer (Sep 2023)

IBM — RDBMS (Nov 2022) • Introduction to Python (Aug 2022)

NASSCOM Foundation — Data Analytics Internship Certification